

Aaron Tran

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Education

University of Waterloo

Waterloo, ON

Bachelor of Computer Science

Sep. 2026 – Apr. 2031

- Relevant Coursework: Designing Functional Programs (CS 135), Algorithm Design & Data Abstraction (CS 136), Algebra (MATH 135), Linear Algebra (MATH 136), Calculus 1 (MATH 137)

Experience

Information Communication Technology Intern

Toronto, ON

Toronto Catholic District School Board

Mar. 2025 – May 2025

- Developed an automated file-renaming program in Python to convert VEX Robotics file formats to STL for 3D modeling and editing workflows, making conversions 15.6x faster than the manual process
- Edited video content and collaborated closely with the ICT department head at St. John Paul II, who led the school's Samsung Solve for Tomorrow initiative, helping the team place in the top 12 regional finalists out of 1,000+ schools nationally in Canada

Projects

Retermina § | *Tauri, Rust, React, TypeScript*

2026 – Present

- Built a desktop terminal workspace on Tauri v2 with a Rust backend driving native PTY shells, running fully local with no cloud dependency or subscription
- Engineered a draggable multi-panel layout with react-grid-layout: split terminals, syntax-highlighted editor, file explorer, live project-wide git diff, localhost tracker, and native preview
- Embedded a Claude Code CLI with per-project token tracking, plus Iris, a context-aware command bar that surfaces git, npm, and shell macros based on live repository state

MarryMap § | *Next.js, React, TypeScript, Supabase, Chrome Extension*

2026

1st Place out of 50+ teams – Cursor Hackathon Toronto

- Built an in-app Vendor CRM tracking every vendor through a full sales pipeline (Lead → Contacted → Quoted → Booked) with quote and deposit amounts, due dates, a per-vendor communication log, and CSV export
- Shipped a collaborative dashboard where couples swipe together on vendor options, with automatic matching, shareable invites, and a contact drawer surfacing the full quote and email history per vendor
- Built a Chrome MV3 extension backed by Supabase/Postgres that lets guests clip vendor recommendations directly into the couple's CRM pipeline

nutri.ai § | *Next.js, React, TypeScript, Gemini/Vertex AI, Tailwind CSS*

2026

Best Usage of Gemini Tools (Main Track) out of 20+ teams – Hack the Valley

- Built an AI-powered pregnancy food-safety and nutrition assistant that fuses barcode, photo, and screenshot analysis with a curated 26-source public-health guidance corpus
- Architected a Gemini-powered analysis pipeline with Zod-validated structured outputs, a deterministic severity-floor rule layer, and citation enforcement to block unsupported or downgraded safety verdicts
- Implemented multimodal food identification via barcode scanning, meal photo analysis, and multi-product screenshot extraction, integrated with the Open Food Facts API for real-time product data

LTN (Laser Tag Now) § | *Swift, ARKit, Nearby Interaction, Network.framework*

2026

2nd Place out of 35 teams – SummerHacks

- Built a peer-to-peer iOS laser tag game using Nearby Interaction (Ultra-Wideband) and Network.framework for direct phone-to-phone real-time gameplay, with no backend server or venue Wi-Fi required
- Engineered a UWB aiming system combining distance/direction ranging with ARKit camera-assisted bearing to resolve hits within a 9° targeting cone
- Built a Bonjour and direct-TCP peer-to-peer networking layer with scoped lobby discovery and a Codable wire protocol syncing game state across up to 8 players

Technical Skills

Languages: TypeScript, Python, Rust, Swift, JavaScript, HTML/CSS

Frameworks & Libraries: React, Next.js, Node.js, Tauri, Tailwind CSS, shadcn/ui, ARKit, Network.framework

Tools & Platforms: Git, MongoDB, Supabase, Vercel, Vultr

AI / ML: Gemini/Vertex AI